

WESTOVER AFB, MA, USA

WMO: 744910

Lat: 42.200N

Lon: 72.533W

Elev: 241

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14703

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	4.3	-11.6	3.0	2.9	-6.8	3.9	8.6	26.8	27.6	24.5	25.9	4.4	40	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	90.7	73.3	87.8	71.6	85.1	70.2	76.2	85.8	74.7	83.2	73.2	80.7	10.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	125.5	79.6	72.2	120.2	78.6	70.7	114.1	77.5	39.6	85.8	38.2	83.1	36.9	81.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.6	19.4	17.3	-12.8	96.3	6.5	2.8	-17.5	98.3	-21.3	100.0	-25.0	101.5	-29.7	103.6		
			-13.2	79.2	6.4	1.4	-17.8	80.2	-21.6	81.1	-25.2	81.9	-29.8	82.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	24.5	26.4	36.0	47.8	58.6	66.9	73.0	70.3	63.0	51.9	40.6	31.3		
	DBStd	18.59	10.92	11.08	9.60	8.66	7.78	6.16	5.29	5.25	7.44	7.93	8.66	9.37		
	HDD50	3012	792	663	445	142	12	0	0	0	0	71	304	583		
	HDD65	6369	1257	1081	898	520	232	49	4	13	127	412	731	1045		
	CDD50	2765	1	2	11	76	278	507	714	629	392	129	23	3		
	CDD65	648	0	0	0	4	34	105	253	177	69	6	0	0		
	CDH74	6427	0	0	6	88	476	1109	2499	1619	588	42	0	0		
	CDH80	2132	0	0	0	26	154	335	956	490	167	4	0	0		
	WSAvg	6.8	7.6	8.1	8.4	8.1	6.8	5.9	5.5	4.9	5.4	6.2	7.0	7.2		
	PrecAvg	46.3	3.1	2.8	3.9	3.6	3.7	4.3	4.1	4.5	4.4	4.3	3.4	3.9		
Precipitation	PrecMax	65.3	6.6	9.3	7.0	7.5	7.0	10.1	10.0	10.0	10.4	13.8	5.8	6.9		
	PrecMin	32.0	1.3	1.4	0.7	1.0	1.1	1.7	2.0	1.2	1.2	0.5	0.6	0.8		
	PrecStd	8.2	1.2	1.5	1.6	1.6	1.5	2.3	2.2	2.3	2.8	2.8	1.2	1.4		
	PrecStd	8.2	1.2	1.5	1.6	1.6	1.5	2.3	2.2	2.3	2.8	2.8	1.2	1.4		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.9	61.1	71.0	83.9	89.7	92.7	95.7	91.9	90.1	78.6	68.7	61.9		
		MCWB	56.5	53.5	56.3	64.0	69.2	73.8	76.4	74.6	73.8	65.6	58.6	57.6		
	2%	DB	50.4	52.3	62.1	74.8	84.5	87.4	91.3	87.9	84.8	73.4	63.8	55.1		
		MCWB	46.5	43.8	50.2	57.9	67.4	70.8	74.0	72.4	70.6	63.3	57.3	51.0		
	5%	DB	43.8	46.3	56.2	69.0	79.3	83.6	88.6	85.3	80.6	69.8	60.0	50.2		
		MCWB	40.0	40.3	46.5	54.7	63.6	68.0	72.5	70.7	68.7	62.4	53.7	46.2		
	10%	DB	39.2	41.5	51.1	63.8	74.6	80.5	85.6	82.5	76.8	66.1	55.3	45.6		
		MCWB	35.0	36.7	43.9	52.6	60.8	66.5	70.9	69.6	66.8	58.7	48.5	41.9		
	0.4%	WB	56.9	56.1	59.4	65.2	73.2	76.1	78.4	77.3	75.8	70.7	62.9	58.4		
		MCDB	58.1	60.0	67.6	75.8	85.6	87.9	90.1	87.1	84.9	74.0	65.3	59.8		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.2	46.5	53.1	62.3	69.7	74.0	76.7	75.2	73.3	67.0	59.1	52.3		
		MCDB	48.3	50.4	59.0	71.1	79.0	83.7	86.9	83.0	80.2	70.7	62.2	54.7		
	5%	WB	40.1	40.8	48.1	58.0	67.2	72.0	75.2	73.5	71.3	63.9	54.5	46.7		
		MCDB	43.4	45.1	54.2	65.1	74.6	79.3	84.0	80.6	77.7	68.5	59.2	49.3		
	10%	WB	35.6	37.0	44.1	53.7	64.0	69.5	73.6	71.9	69.0	60.1	50.0	42.0		
		MCDB	38.4	41.1	50.2	62.7	71.6	76.2	81.3	79.1	74.4	64.4	54.5	45.0		
Mean Daily Temperature Range	5% DB	MDBR	17.7	19.5	19.7	23.0	23.7	22.3	22.3	22.9	23.0	22.0	19.3	16.2		
		MCDBR	20.2	24.7	28.7	33.0	31.8	27.2	25.8	25.6	26.0	26.2	24.3	21.4		
		MCWBR	17.7	18.2	17.5	17.3	15.3	12.7	10.4	11.6	13.2	15.8	17.7	18.2		
	5% WB	MCDBR	19.4	22.4	24.5	26.5	25.9	22.3	21.5	20.8	21.0	20.6	20.2	19.8		
		MCWBR	18.0	17.8	16.5	16.9	14.3	12.1	9.8	10.4	12.0	14.7	17.0	19.0		
Clear-Sky Solar Irradiance	taub		0.292	0.306	0.331	0.373	0.410	0.457	0.474	0.443	0.381	0.341	0.311	0.293		
	taud		2.441	2.411	2.431	2.357	2.289	2.189	2.150	2.224	2.375	2.501	2.543	2.527		
	Ebn at Noon		272	285	291	284	275	261	255	259	269	270	264	263		
	Edh at Noon		24	29	32	37	41	46	47	43	34	27	22	21		
All-Sky Solar Radiation	RadAvg		563	833	1161	1481	1643	1779	1856	1659	1359	882	616	460		
	RadStd		43	73	83	159	179	174	111	88	101	89	53	46		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+88

Nomenclature: See separate page